CARRT Assistive Arm CSE Project — Weekly Status Report # 3

Current Report Period: Weeks #1-3 ***(End-of-Period Report)***

1. Goals for the Current Period?

Carry-Over Goals (unaccomplished from prior weeks/reporting periods)

a. **None, first week.**

New Goals

1. **ADL Task Selection –** [✓ | ✗] – Select and document three Activities of Daily Living (ADL) tasks to be performed by the Kinova Gen3 arm.
   a. ***Approach:*** Hold a team meeting to evaluate the feasibility and relevance of different possible tasks, as well as testability. Finalize and document the three tasks, tie them to user stories and insert them into project requirements.
2. **ROS2 Installation –** [✓ | ✗] – Complete installation of ROS2 and verify functionality by running documented sample publisher-subscriber nodes (ROS2 guides).
   a. ***Approach:*** Follow official ROS2 installation documentation on Ubuntu. Verify by running provided sample nodes in Ubuntu and VS Code. All members gain access to a GitHub workspace for coding purposes/environment creation.
3. **Gazebo Installation / Simulation –** [✓ | ✗] – Complete install of Gazebo simulator and verify by simulating the Kenova Gen3 Robotic Arm performing at least one task in Gazebo.
   a. ***Approach:*** Install Gazebo alongside ROS2. Load and verify the arm's URDF model from Kinova Kortex packages. Attempt to run at least one ADL simulation sequence to check viability.

2. Goals Accomplished in the Period?

1. ***Task #1:*** **ADL Task Selection –** [**Accomplished** / Partially Accomplished]: Tasks Selected: (1) Clear a table of 3+ objects, (2) Pick an object off the floor, (3) Flip a light switch on/off. Chosen to align with the user stories chosen in the Week 2 company contact meeting.
   a. ***Completed By: Entire Team (~15-30 min each)***
2. ***Task #2: ROS2 Installation –*** [**Accomplished** / Partially Accomplished]: ROS2 was successfully installed and verified on Ubuntu using the Humble distribution. The sample nodes at the end of tutorials for the installation ran without error.
   a. Completed By: Oliver (3 hrs), Jadon (3 hrs), Allison (2 hrs), Brielle (2 hrs)
3. ***Task #3: Gazebo Installation –*** [Accomplished / **Partially Accomplished**]: Gazebo was successfully installed and the Kinova Gen3 arm was rendered. The arm URDF loaded correctly, and the Gazebo environment was functional. Joint actuation / arm movements were not yet achieved before the end of the period.
   a. Completed By: Oliver (3hrs), Jadon (3hrs)

3. Critical Reflection on Unaccomplished Goals for the Period (to Carry-Over)

1. ***Task #3:*** **Gazebo Simulation –** The team was unable to achieve arm movement in Gazebo simulator due to insufficient familiarity with the simulator's actuation and control configuration. Time was mostly spent on installation, troubleshooting, and model loading, with little time left to experiment with movement interfaces.
   a. ***Completed By: Oliver (2 hrs)***
   b. Future / Revised Approach: Redistribute Gazebo simulation work over other team members. Research ROS2 Gazebo and its plugins. Work towards achieving arm movement within Week 4 to move, then move to task scripting.

4. Major Goals Remaining for Current Period –OR– Major Goal List for Next Period (End of Period)

1. **Gazebo ADL Simulation –** Successfully test at least 2 of 3 ADLs using the simulated arm in Gazebo, with observable arm movement and task completion.
   a. ***Approach:*** Build using the Gazebo model. Implement joint control scripts with ROS2. Run and verify each ADL pose sequence (floor/table/light switch).
2. **UI Development –** Begin development of UI for the arm and evaluate methods for a tablet interface for user interaction.
   a. ***Approach:*** Research ROS2-native UI frameworks and iPad mirroring/display options. Compare at least two options of implementation.
3. **QR Code Recognition Planning –** Determine an object recognition strategy using QR codes (AprilTag and OpenCV)
   a. ***Approach:*** Evaluate AprilTag vs. other QR approaches. Define a finalized object list and QR placement rules. Produce example implementation.

5. Work Division for the Team

- *Divisions are specifically noted in Sections 2 and 3. For each team member, by total hour count for the current week:*

Name:	Oliver Gaston	Jadon Orr	Allison Moline	Brielle Ashmeade
Hours Total:	~8 hrs	~6.5 hrs	~2.5 hrs	~2.5 hrs

Format designed to manage both course-assigned and weekly status reports. To detail and track progress on major goals across report periods, noting what was completed and what carries forward to next week or report period, to maintain understanding of current project needs.